

Part 18: Hypothesis Testing

Summary

Statistical hypothesis testing is a formal framework for “making discoveries.” It is ubiquitous in research, and overused. Nevertheless, it can be useful, and it is definitely important to understand the concept.

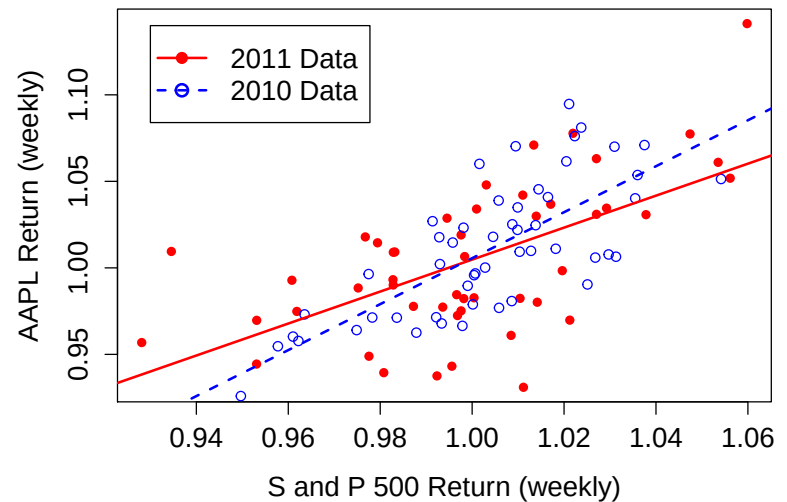
In particular, one will see repeated references to **p-values** when reading research. These are often misinterpreted.

Broadly speaking, hypothesis testing gives us a basis to make a claim that an observed “feature” is “real” and not simply an artifact that occurred due to random variability in our sample.

Consider the questions on the following three slides. In each case, the question is not asking for the best estimate of a parameter, but instead asking if the data provide sufficient evidence for some claim.

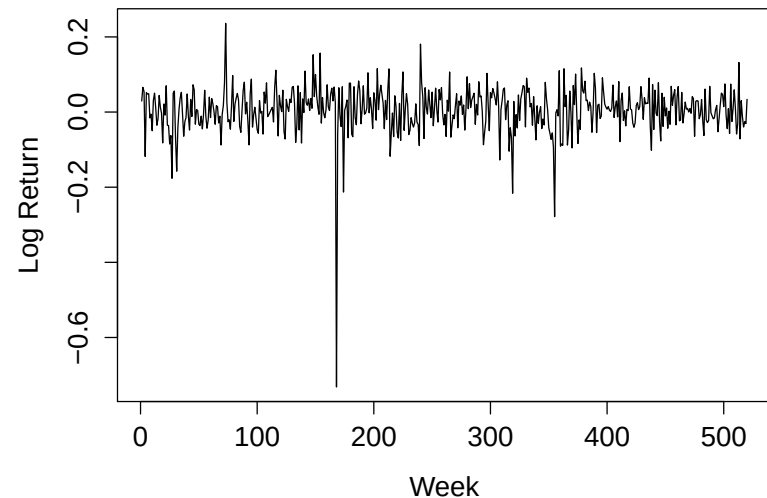
What do these examples have in common?

1. They each ask about a feature of the **population** from which the **sample** (observed data) is drawn.
2. The answer to each question is definitely “No.” The issue is whether or not the data provide **sufficient evidence** that the answer is “No.”
3. Each question puts forth a relatively simple “null hypothesis” and asks “Is there sufficient evidence to reject the null?”



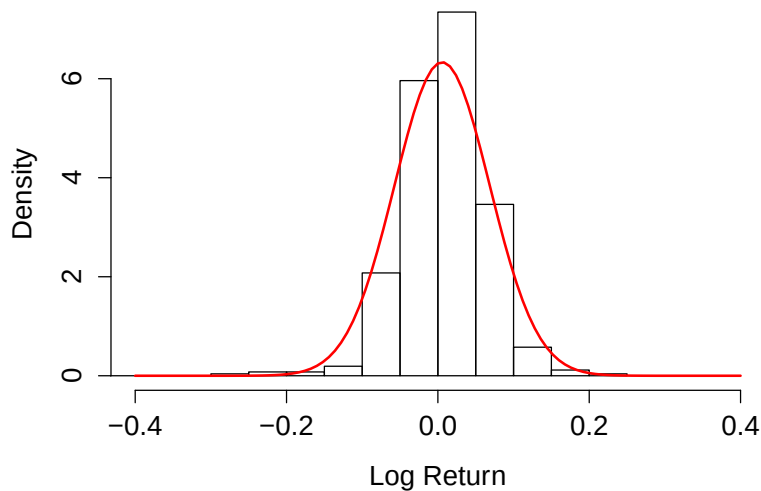
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2 Is the slope of the regression line the same in 2010 and 2011?



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2 Is the volatility constant over this time period?



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2 Are these data normally distributed?

1 Such questions as these can be addressed using statistical hypothesis testing.

3 The key requirement is that the question of interest, "Is there sufficient evidence that...," can be turned into a question regarding the population.

5 Often the hypothesis will take the form of a question regarding θ , but this is not always the case.

1 Null and Alternative Hypotheses

2 The first two key ingredients to a hypothesis test are the **null hypothesis**
3 (H_0) and the **alternative hypothesis** (H_a or H_1).

4 **The alternative hypothesis is the claim you are testing to see if there is**
5 **strong enough evidence to support.**

6 The null hypothesis is the "baseline" or "status quo."

7 The main thing to keep in mind is the following: **Statistical hypothesis**
8 **tests are designed to determine if there is strong evidence to support the**
9 **alternative (and reject the null). They are not designed to test if there is**
10 **strong evidence to support the null hypothesis.**

1 **Exercise:** A pharmaceutical company is testing a new treatment for a dis-
2 ease. The current treatment is 40% effective. Explain why a hypothesis
3 test is appropriate in this case. Formulate a reasonable null and alternative
4 hypothesis for the company.

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- 1 **Exercise:** Write out null and alternative hypotheses for each of the three
2 original examples. Be clear about what is being tested in each case.

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1 **The Test Statistic and the Rejection Region**

- 2 Not surprisingly, data are used to test the hypothesis. As before, random
3 variables $X_1, X_2, \dots, X_n$ are used to model the observable sample.

- 4 The **test statistic** is the function of the data that we will base our conclusion
5 on. Keep in mind that the test statistic is also a random variable.

- 6 Write $T = T(X_1, X_2, \dots, X_n)$.

- 7 The **rejection region** is the set of values for the test statistic such that the
8 conclusion is “reject the null hypothesis in favor of the alternative.” In
9 other words, when T falls into the rejection region, we conclude that there
10 is strong evidence in support of the alternative.

- 1 *How is the rejection region chosen?*

- 2 Consider the two possible errors we can make with a hypothesis test:

- 3 1. A **Type I error** is made when we reject the null even though the null is
4 true.

- 5 2. A **Type II error** is made when we fail to reject the null when the alter-
6 native is true.

- 7 The assumption is that we are **primarily concerned with making a Type I**
8 **error**. We want to limit the probability of making a Type I error.

- 9 We form the rejection region so that the probability of a Type I error is less
10 than or equal to some chosen amount α . This is often called the “size” or
11 “level” of the test.

- 1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$, with σ^2
2 known. We want to test $H_0 : \mu = 10$ versus $H_1 : \mu > 10$. A natural test
3 statistic in this case is $\bar{X}$. Formulate the rejection region for a level α test.

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1 In the previous exercise, we note that the result was based on knowledge
2 of the distribution of the test statistic under the null hypothesis. It is key
3 that the test statistic be chosen so that its distribution is known (at least
4 approximately), under the null. As usual, the central limit theorem can be
5 very helpful.

6 Also, we note that in the exercise we tested the **one-sided alternative hy-**
7 **pothesis** $H_1: \mu > 10$. In other cases we will test a **two-sided alternative**, for
8 example, $H_1: \mu \neq 10$. In such a case the null would be rejected for either
9 small or large values of the test statistic.

1 **Exercise:** Suppose that we tested $H_1: \mu \neq 10$ in the above example. What
2 would the rejection region be for fixed α ?

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1 The p-value

2 The choice of α in the statistical hypothesis testing framework may seem
3 arbitrary. Why would one choose one value of α over another?

4 As α is chosen smaller, the experimenter is choosing to be more cautious,
5 to protect against Type I error. The opposite is true as α is chosen larger.

6 For a given data set, there is a **dividing line** between the choices of α for
7 which the conclusion is “reject the null” and the conclusion is “fail to reject
8 the null.” This dividing line is the **p-value**.

9 1. If $\alpha < \text{p-value}$, then “fail to reject the null.”

10 2. If $\alpha > \text{p-value}$, then “reject the null.”

1 It is important to note that the p-value is **a function of the data**.

2 The p-value is **not** the probability that the null hypothesis is true. Instead,
3 it is the largest choice of α such that the conclusion is “fail to reject the
4 null.”

5 The appeal is that one can report a p-value, and allow the reader to decide.
6 Most software packages report the results of hypothesis tests as p-values.

7 The next exercise illustrates another interpretation of the p-value: It is
8 the probability that, assuming the null hypothesis is true, the test statis-
9 tic would be equal to or more extreme in favor of the alternative than the
10 observed test statistic.

11 This interpretation is often useful for making computations.

1 **Exercise:** Return to our example when testing $H_0: \mu = 10$ versus $H_1: \mu >$
2 10. Assume that $\sigma^2 = 1$. We observe $\bar{x} = 10.5$ with a sample size of $n = 50$.
3 What is the p-value in this case? What happens if we switch to $H_1: \mu \neq 10$?

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1 The Wald Test

2 The **Wald Test** is built around a simple notion: An ideal choice for the test
3 statistic is the number of standard errors a parameter estimate is from the
4 null hypothesized value for that parameter.

5 Specifically, if testing $H_0: \theta = \theta_0$, then the test statistic is chosen as

$$T = \frac{\hat{\theta} - \theta_0}{SE(\hat{\theta})}$$

7 T gives the “the number of standard errors $\hat{\theta}$ is from θ_0 .”

8 In a wide range of situations, T has approximately the standard normal
9 distribution under the null hypothesis. Hence, the region region is not
10 difficult to derive.

1 **Exercise:** What is one very important case in which we can state that T is
2 approximately standard normal under the null?

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1 The three possible tests can be presented as follows:

2 1. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta > \theta_0$, then reject the null if $T > z_\alpha$.

3 2. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta < \theta_0$, then reject the null if $T < z_\alpha$.

4 3. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta \neq \theta_0$, then reject the null if either
5 $T < z_{\alpha/2}$ or $T > z_{\alpha/2}$.

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. Bernoulli(p). Construct the
2 two-sided Wald Test for p . Suppose that $n = 100$ and $\sum X_i = 47$. Test
3 $H_0: p = 0.4$ versus $H_1: p \neq 0.4$. What is the p-value?

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1 The Wald Test can be extended to testing multiple parameters simultane-
2 ously by noting the following result: If $\mathbf{X}$ is p -dimensional multivariate
3 normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, then

$$4 \quad (\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})$$

5 has the chi-squared distribution with p degrees of freedom.

6 **Exercise:** Explain how this result can be used in conjunction with a multi-
7 parameter distribution when the MLE is used as the estimator.

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